

Practice Questions

Question #1

A computer consulting firm presently has bids out on three projects. Let A_i denote the event that project i is awarded, for $i = 1, 2, 3$, and suppose that $P(A_1) = 0.2$, $P(A_2) = 0.3$, $P(A_3) = 0.4$, $P(A_1 \cap A_2) = 0.1$, $P(A_1 \cap A_3) = 0.1$, $P(A_2 \cap A_3) = 0.1$, and $P(A_1 \cap A_2 \cap A_3) = 0.05$. Express in words each of the following events, and compute the probability of each event:

- $A_1 \cap A_2$
- $A_1 \cup A_2$
- $A_1 \cap A_2 \cap A_3$
- $A_1 \cup A_2 \cup A_3$
- $A_1 \cap A_2 \cap A_3^c$
- $A_1 \cap A_2^c \cap A_3^c$

Question # 2

Suppose that 55% of all adults regularly consume coffee, 45% regularly consume carbonated soda, and 70% regularly consume at least one of these two products.

- What is the probability that a randomly selected adult regularly consumes both coffee and soda?
- What is the probability that a randomly selected adult doesn't regularly consume at least one of these two products?

Question # 3

The computers of six faculty members in a certain department are to be replaced. Two of the faculty members have selected laptop machines and the other four have chosen desktop machines. Suppose that only two of the setups can be done on a particular day, and the two computers to be set up are randomly selected from the six (implying 15 equally likely outcomes; if the computers are numbered 1, 2, ..., 6, then one outcome consists of computers 1 and 2, another consists of computers 1 and 3, and so on).

- What is the probability that both selected setups are for laptop computers?
- What is the probability that both selected setups are desktop machines?
- What is the probability that at least one selected setup is for a desktop computer?
- What is the probability that at least one computer of each type is chosen for setup?

Question # 4

One box contains six red balls and four green balls, and a second box contains seven red balls and three green balls. A ball is randomly chosen from the first box and placed in the second box. Then a ball is randomly selected from the second box and placed in the first box.

- What is the probability that a red ball is selected from the first box and a red ball is selected from the second box?
- At the conclusion of the selection process, what is the probability that the numbers of red and green balls in the first box are identical to the numbers at the beginning?

Question # 5

A certain shop repairs both audio and video components. Let A denote the event that the next component brought in for repair is an audio component, and let B be the event that the next component is a compact disc player (so the event B is contained in A). Suppose that $P(A) = 0.6$ and $P(B) = 0.2$. What is $P(A \cap B)$?

Question # 6

At a certain gas station, 40% of the customers use regular gas (R), 35% use plus gas (P), and 25% use premium (PR). Of those customers using regular gas, only 30% fill their tanks (event B). Of those customers using plus, 60% fill their tanks, whereas of those using premium, 50% fill their tanks.

- What is the probability that the next customer will request plus gas and fill the tank (P ∩ B)?
- What is the probability that the next customer fills the tank?
- If the next customer fills the tank, what is the probability that regular gas is requested? Plus? Premium?

Question # 7

Seventy percent of the light aircraft that disappear while in flight in a certain country are subsequently discovered. Of the aircraft that are discovered, 60% have an emergency locator, whereas 90% of the aircraft not discovered do not have such a locator. Suppose a light aircraft has disappeared.

- If it has an emergency locator, what is the probability that it will not be discovered?
- If it does not have an emergency locator, what is the probability that it will be discovered?

Question # 8

An oil exploration company currently has two active projects, one in Asia and the other in Europe. Let A be the event that the Asian project is successful and B be the event that the European project is successful. Suppose that A and B are independent events with $P(A) = 0.4$ and $P(B) = 0.5$.

- If the Asian project is not successful, what is the probability that the European project is also not successful? Explain your reasoning.
- What is the probability that at least one of the two projects will be successful?
- Given that at least one of the two projects is successful, what is the probability that only the Asian project is successful?

Question # 9

An aircraft seam requires 25 rivets. The seam will have to be reworked if any of these rivets is defective. Suppose rivets are defective independently of one another, each with the same probability.

- If 20% of all seams need reworking, what is the probability that a rivet is defective?
- How small should the probability of a defective rivet be to ensure that only 10% of all seams need reworking?

Question # 10

A boiler has five identical relief valves. The probability that any particular valve will open on demand is .95. Assuming independent operation of the valves, calculate $P(\text{at least one valve opens})$ and $P(\text{at least one valve fails to open})$.

Question # 11

- a. A lumber company has just taken delivery on a lot of boards. Suppose that 20% of these boards (2,000) are actually too green to be used in first-quality construction. Two boards are selected at random, one after the other. Let A and B . Compute $P(A)$, $P(B)$, and $P(A \cap B)$ (a tree diagram might help). Are A and B independent?
- b. With A and B independent and $P(A) = P(B)$, what is $P(A \cup B)$? How much difference is there between this answer and $P(A \cup B)$ in part (a)? For purposes of calculating $P(A \cup B)$, can we assume that A and B of part (a) are independent to obtain essentially the correct probability?
- c. Suppose the lot consists of ten boards, of which two are green. Does the assumption of independence now yield approximately the correct answer for $P(A \cup B)$? What is the critical difference between the situation here and that of part (a)? When do you think an independence assumption would be valid in obtaining an approximately correct answer to $P(A \cup B)$?

